Using Data Analytics and Visualization Dashboard for Engineering, Procurement, and Construction Project's Performance Assessment

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***Abstract*—This study demonstrates the application of design principles for tools in Engineering, Procurement, and Construction (EPC) projects for project management purposes. It advocates the use of proper data analytics and visualization that can be implemented to support effective project progress reporting as well as performance monitoring.**

At first, an Entity Relationship Diagram (ERD) of the collected data was developed, and then the database was retrieved into the Microsoft Power BI for analysis and visualization. The project's relevant details were visualized and analyzed in terms of the major key performance indications that help evaluate the current situation of projects and aids in future decision making for project performance and portfolio management. A real case of a construction company has been selected and examined. Analytical results support finding the story behind the data.

On the other hand, the effects point out that the use of the suitable facts analytics method coupled with the right analytics method and appropriate data visualization software would result in optimum use of information for future aspiration of project success and proper project progress reporting and performance evaluation. It will help companies to transfer from traditional data storage style to big data analytics and powerful use of enterprise business data for companies' growth and success in the field of EPC projects and the construction industries as a whole. Furthermore, it can be used for quicker and extra decisive choices aiming to keep projects on music about their security performance, scheduled time, cost, and great level.

Using the proposed dashboard is creating a summary of the accessible records that prints a photo of how initiatives and portfolios are performing, permitting decision-makers to take their future strategic steps aiming for the improvement of their initiatives and accordingly success in their cutting-edge and future endeavors to reap the objective of all stakeholders of this organization. In this paper, it was demonstrated that the visualization of the contractor's performance and KPI are bringing assurance on contractor performance in addition to daily operational monitoring. Furthermore, it helps managers in organizing the workload to ensure the project's completion timely and meeting the customer demand as expected.

Keywords-data analytics; dashboard; entity relationship diagram; portfolio management; power BI

I. Introduction

Data is of high importance to any organization. The waft of data within an organization is key to its success. [1]. It is a challenging task for organizations to collect data and retrieve needed information as well as making the right decisions based on the extracted information alone. Therefore, many systems have been created to make the decision-making process faster and easier to a certain level. However, if such systems are not used properly (or sometimes not used at all), they can avert and hinder the decision-making process. Since decisions in any company are made solely by people (employees, mid-management, and senior management), therefore the way records and visualized play a key position in the procedure of decision making [2].

As the value of information and the way it's understood, controlled and used for performance improvement is being realized and asserted, it is important to know what should be measured and recorded as information in order to bring benefit from such information [2-4]. This will help discern what is meaningful to measure and how well teams are aligned to performance objectives. It is advised that teams leverage existing data and conventional metrics [5-6].

Proper data analysis and associated tools provide a means of data manipulation for the available data warehouse to aid in proper decision-making. This data must be of a high level of accuracy so that statistical correlation and the produced analysis is correct to have a sound predicting trends and proper decision to be taken [7]. This is evident in any business but particular construction companies [5-6]. These analytics techniques and associated methods and tools are considered important tools for improving project results, proper decision making for future projects as well as reducing risk. By empowering construction companies to leverage the huge amounts of data they already collect, analytics can harness important insights that allow the quality of management decisions to be faster, more effective, and much improved. Such data analytics can help project teams assess advanced progress reporting, portfolio management, and targeted project performance improvement [5-6].

The main aim of this novel research is to develop a new tool that utilizes and visualizes the existing data to provide a representative status of the projects as well as to allow decision-makers in targeting their goals and increasing the probability of future project success. Furthermore, to showcase the benefits of introducing pioneering data analytics and techniques in construction companies.

The proposed tool is providing sufficient solutions for the project portfolio managers in monitoring the performance indicators, verify the availability of resources, meet customer satisfaction, and validate contractor overall performance.

A. Using Dashboards for Construction Projects

Dashboards for data visualization through the use of applications labeled as dashboard interfaces enable several users throughout the organization to make sound decisions in a timely manner by merging and evaluating a wide array of data [8]. For construction projects, [9] provides a definition of a performance dashboard as "a multilayered software built on enterprise intelligence and data integration infrastructure that allows agencies to measure, monitor, and manage overall performance more correctly" [10]. Dashboards can be used to deliver important points about the overall performance of portfolios of development initiatives in a nice manner [11]. Dashboards have been used in the construction industry for progress reporting and performance evaluation. The authors recommended a method for effective schedule and budget performance for construction portfolios by proposing the use of dashboards. Such a system was used for portfolio performance assessment [11].

There are numerous benefits of using dashboards in construction industries such as effective decision-making; increased productivity; project successes and failures discussing of likelihood; manual reporting cost-saving; use it as database system to store data for projects in hand (serve as historical data for projects of similar nature); portfolio management; and contractors performance monitoring and management [12-14].

B. Selecting KPI's in EPC Projects Dashboards

Before deciding on the proper analytics techniques, methods, and tools to be used for data analysis and visualization, Key Performance Indicators (KPIs) have to be identified and used inside suitable dashboards to permit the perfect administration of a portfolio of construction projects. These KPIs are chosen from key effects areas that exhibit the importance of key records gathered from different projects [13-15].

Recognizing what KPIs for EPC projects are to be displayed on the dashboard is a highly important undertaking since there are more than a few corporations and stakeholders in one company that have a situation about the usual repute of a portfolio [16-17].

C. Data Analytics and Techniques

For progress reporting and performance monitoring in construction projects in general, the most suitable data analytics technique to be used is descriptive analytics. Descriptive analytics answers the simple question "what happened," as it reports the last and current data of the projects [3], [18]. It is considered as a backward view of information to demonstrate what took place in the past.

Descriptive analytics is a technique that utilizes historical data to recognize relationships, patterns. it is concerned with modeling of past performance [18]. Descriptive analytics examines data and information to define the present business condition in a mode that developments, relationships, and exclusions become apparent. This shall be coupled with the appropriate data visualization tool in order to find the ideal match that provides a summary of what happened in order to be able to have better progress reporting and performance gaging of the project in hand [3].

D. Data Visualization Tools

Visualizing the data is the data arrangement of a presentable manner in a graphical form. To analyze and get improved perceptions from the data stored in a company, it has to be epitomized in a methodical format [2].

Data visualization can be used as a technique to identify areas that need attention or improvement, provide sound progress reporting, and ultimately guide future decision making and performance improvement. The data visualization process, according to [2], consists of the flowing steps in Fig. 1:

Figure 1. Flow chart of data visualization

One of the Data visualization software is Power BI, which is affiliated to Microsoft to provide business analytics services. It offers interactive visualization as well as self-service commercial enterprise intelligence.

E. Power BI

Power BI is an enterprise analytics solution that lets businesses visualize their statistics and share insights across their corporation or embed them in their app or internet site [19]. Power BI permits users to connect to heaps of statistics sources and deliver the agency's records to life with stay dashboards and reports. When it comes to data, the faster a company can make sense of the data it has in the organization, the smarter and faster it will aid in its business advancement. Microsoft Power BI offers a means of connecting data, prepping it, model, and visualize it in order to share insights. It helps to connect to thousands of databases on the premises or via the cloud. It allows turning tedious excel sheet information into real-time visual information and turning data into life interactive visuals. Through Microsoft Power BI, one can create dashboards that deliver 360 degrees of any business aspect, as it is free and very user-friendly. It truly helps to transfer data into smarter decisions for businesses [19].

II. RESEARCH METHODOLOGY

A. Data Collection

In this research, secondary data from 13 different EPC projects were collected from a construction leading company in Qatar. Due to information confidentiality, the company's name can't be disclosed. The collected data covered all general information and KPIs used in the company for its regular reporting of EPC project progress. The company reported that for effective data reporting, they designated 5 major categories of information, their related subcategories, and KPIs, which are listed in Table I.

The obtained details were gathered from 13 projects of different nature, monetary value, and location as it ranged from traditional building projects to infrastructure and train-related work owned or operated under the authority of the company. It has been reported by a senior manager in this construction company that project data were not employed in a visual manner nor a proper analytical technique. The traditional progress reporting mainly utilized the traditional use of Microsoft Excel, Microsoft Word, and PowerPoint software. Such tools that are currently being used by the company did not enable the projects department to perform proper analysis or provide a proper visual output that could assist in a better decision-making process.

In this analysis, it was deemed that the most suitable analytics technique to be employed is descriptive analytics.

TABLE I. PROJECT DETAILS MAIN CATEGORIES AND SUB-CATEGORIES

Sr.	*Major Category*	*Sub-Categories*
1	Project Information	Project Title, Cost, Complexity, Category, and Expected Completion
2	Project Area	Project Purpose, Location, Area No. and Street No.
3	Assigned PMT[a]	Head of Project, Project Engineer/Manager and Supporting Resources
4	Contract Information	Contracting Strategy, Contractor Name, Engineering Contractor and Suppliers
5	Project Performance (KPIs)	- Schedule Status: Planned, Actual, Variance, and Remarks - Cost Status: Planned, Actual, Variance, and Remarks - Safety Status: No. of Lost Time Incident (LTI) and Remarks - Quality Status: Non-Conformity Report and Remarks - All Status: Overall Performance (OP) and Remarks

B. Analysis Methodology

Entity Relationship Diagram (ERD) is a type of structural diagram for use in database design. ERD is the diagramming design that recognizes and stipulates the type and category of relationships and nodules. ERD's are used to streamline designing management and decision support systems. it reduces time spent on complicated coding routines as well as saving time in documentation processes of any business [20-23]. The ERD is known for its ease of use, as well as strength in real-world problems modeling that can be converted into a database like representation.

Finally, Microsoft Power BI was selected as the designated tool that provides the most up-to-date data visualization tool in terms of developing a project progress-reporting tool. The data was collected, cleansed, and ERD as well as was drawn using the Lucid Chart as well as tools provided in excel linked to Power BI software in order to depict the relationship between the designated entities within the tables.

III. RESULTS AND DISCUSSION

Enterprise ERD has been established for the construction company case in order to easily fill the data and information required prior visualization. For visualizing and picturing the data and analyzing the information, Microsoft Power BI was used, and multiple dashboards have been created. The following figures are captions from the designed dashboard via Power BI. They show the status of the reported projects in hand and their performance under different KPI's that were assigned and prechosen by senior management at the company.

The dashboard has been split into five tabs based on the company's needs, which are: Overall Summary, Contractor Performance, Project Performance, Project Variance, and Expected Completion. Each one of the tabs shows different analyses of the available data warehouse, and it can even be expanded more based on the organization's needs.

Fig. 2 represents the second tab titled "Contractors performance," which is evaluated based on the KPIs set in the company. It shows the ranking of the contractors in each KPI as well as the overall ranking per contractor. This tab also shows other factors influencing these KPIs such as Subcontractors and suppliers assigned for the projects.

Fig. 3 reveals each project's performance as well as the performance of each KPI in the company overall. For example, it shows that in these projects, there were 23 Lost Time Incidents. Also, this part of the dashboard shows the program/portfolio performance per head, Project Manager, and location.

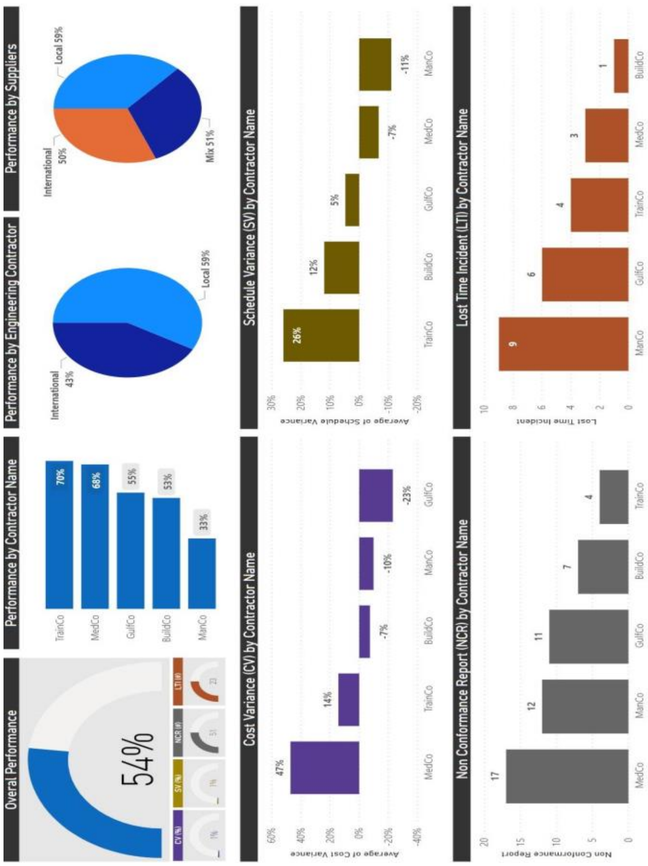

Figure 2. Power BI screenshot contractors performance.

IV. Conclusion and Future Works

In this paper, the concept of data analytics techniques and data visualization dashboards for performance and progress reporting in projects have been introduced to monitor the portfolio's performance. Looking into the business data from different dimensions provides benefits to the project owners and stakeholders for decision making.

The use of Microsoft Power BI along with descriptive data analytics technique and data mining and clustering methods allows management and decision making on these projects to take the right course of action for current and future endeavors. It is advocated to use suitable and advanced business analytics and information tools for tracking several events across EPC projects. This enables the project's leaders to have an ordinary recognition of the project's healthiness and proactively aid decision-makers and stakeholders in projects to make the right decisions based on their right use of business information systems and analytics tools.

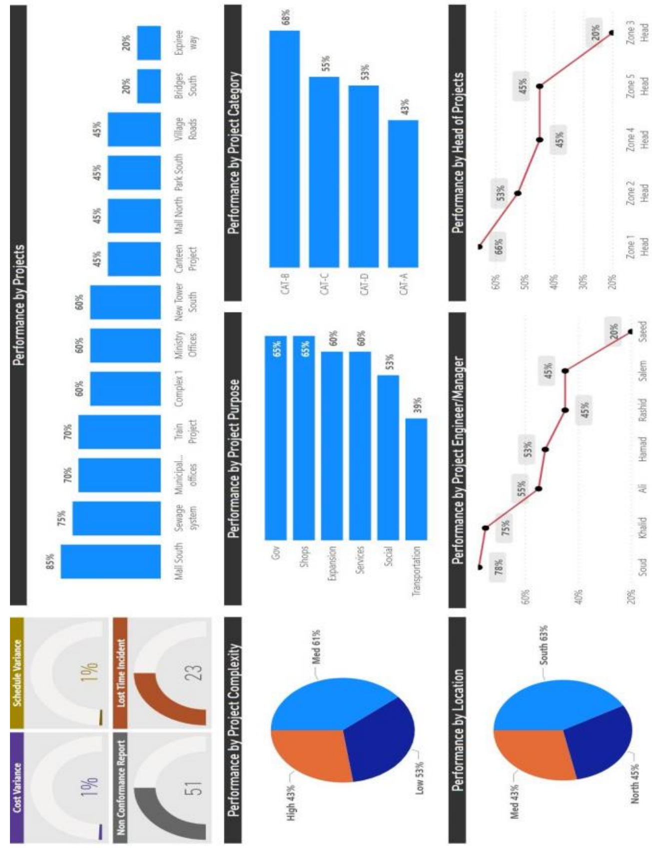

Figure 3. Power BI screenshot projects performance

The outcomes of the construction company case study were gathered and discussed with the project's stakeholders to present the findings of the previous contractor's performance based on several criteria. It was proven that the visualization of the contractor's performance and KPI are assuring contractor selection in addition to day-to-day operational monitoring. Furthermore, it helps managers in organizing the workload considering the bulk number of projects going on at the same time to ensure the delivery of all projects within the timeframe, budget, quality, and compliance manner.

In conclusion, although switching from the traditional progress reporting to the data-driven approach and utilizing the available analytical tools makes perfect sense, the company did not explore the use of an analytical approach, and there might be a sense of discomfort to move away from the current traditional approach of reporting which was used for a long time in the company. There is an expected level of resistance to change in many industries, and construction is one of them. The reluctance to do the switch or the unavailability of training in the field of data analytics and the usefulness of the available tools and technologies is present.

Several social, economic, and environmental sustainability metrics such as safety, job-creating, carbon emission, and air pollution can also be integrated to assess the sustainability of EPC projects and several decision-making methods such as life cycle sustainability assessment, expert-judgment based multi-criteria decision-making method, and system dynamics can be employed for assessing the overall project performance [24-25].

The dashboard can be enhanced in the future through Additional criteria for contractor performance measurement:

- Availability of resources & responsiveness to effective mobilization for Emergency EPC projects.
- Performance on engineering submitting rejection & re-submit engineering documents.
- Effectiveness of subcontractor or vendor management.
- Time is taken/effective for close-out of contracts.
- Response to warranty items.
- Contract changes/claims.

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